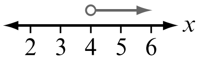
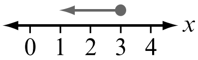
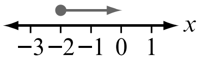
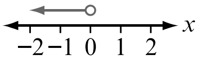
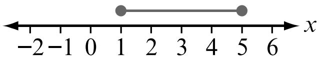
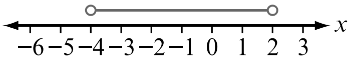
3 a $x > 4$


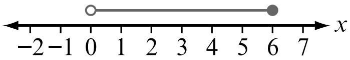
b $x \le 3$


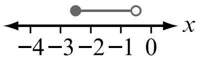
c $x \ge -2$


d $x < 0$


e $1 \le x \le 5$


f $-4 < x < 2$


g $0 < x \le 6$


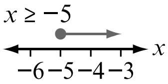
h $-2.5 \le x < -0.5$


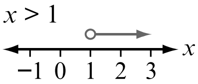
4 a i $3 < 7$; true ii $5 < 9$; true
iii $9 < 13$; true iv $-2 < 2$; true
b i $1 < 5$; true ii $-1 < 3$; true
iii $-5 < -1$; true iv $6 < 10$; true
c i $4 < 12$; true ii $-4 < -12$; not true
iii $6 < 18$; true iv $-2 < -6$; not true
d i $1 < 3$; true ii $-1 < -3$; not true
iii $\frac{2}{3} < 2$; true iv $-2 < -6$; not true
e Inequality sign stays the same when adding or subtracting any number. Also stays the same when multiplying or dividing by a positive number. However, direction of inequality sign is reversed when multiplying or dividing by a negative number.
f Some possible answers are given.
i $4 > -8$, $4 + 2 > -8 + 2$ $(6 > -6)$, $4 - 1 > -8 - 1$ $(3 > -9)$
ii $4 > -8$, $4 \times 2 > -8 \times 2$ $(8 > -16)$, $4 \div 2 > -8 \div 2$ $(2 > -4)$
iii $4 > -8$, $4 \times -2 < -8 \times -2$ $(-8 < 16)$, $4 \div -2 < -8 \div -2$ $(-2 < 16)$

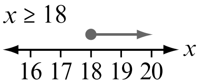
5 a $x + 3 > 8$ b $-2x \ge 8$ c $x \le 10$
d $x < 4$ e $-x \ge -9$ f $x > -16$
g $x \ge 7$ h $-x > 3$ i $-6x > -5$

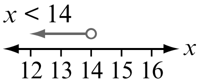
6 a $x < 5$ b $x \ge 7$ c $x \le -6$
d $x > 6$ e $x \le -5$ f $x < 50$
g $x \ge -20$ h $x \le 2$ i $x > -5$

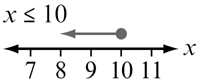
7 a $x \ge -6$ b $x < -2$ c $x \le 3$
d $x > -24$ e $x < -7$ f $x \ge 1$
g $x > -24$ h $x < 7$ i $x \le 10$

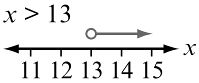
8 a $x \ge -5$


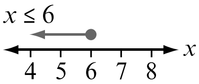
b $x > 1$


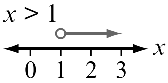
c $x \ge 18$


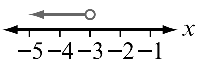
d $x < 14$


e $x \le 10$


f $x > 13$


g $x \le 6$


h $x > 1$


i $x < -3$


9 a $x > 4$ b $x \le -2$ c $x \ge 3$
d $x > 4$ e $x < 1$ f $x \ge -1$
g $x \le 5$ h $x < 2$ i $x > \frac{1}{2}$

10 a $x > 11$ b $x \le 13$ c $x \ge -2$
d $x > -19$ e $x > -15$ f $x \le 1$
g $x \le 3$ h $x > -\frac{1}{4}$ i $x \ge -6.5$

11 a $x \le -7$ b $x > 1$ c $x \le -3$
d $x < 1$ e $x \ge 4$ f $x > -5$

12 a one
b Linear inequalities can have many solutions within a given range.

13 a $p \ge 650\,000$, where p is selling price in \$
b $h > 97$, where h is height of a person in cm
c $s \le 60$, where s is speed in km/h
d $125 \le h \le 196$, where h is height of a person in cm

14 a $3p \le 20$, where p is cost of pack of gum in \$; $p \le 6\frac{2}{3}$
b Todd could buy 1, 2, 3, 4, 5 or 6 packs of gum.

15 a $m = \frac{1}{2}(20 - x)$, where m is number of watermelons they will each take home.
b $\frac{1}{2}(20 - x) \le 3$; $x \ge 14$
c 14 or more watermelons

16 a 25 min b 16 min c 19 min to 29 min

17 a $x \ge -11$ b $x < 4$ c $x \ge 2$ d $x < -8$
e $x \le 2$ f $x \le -3$

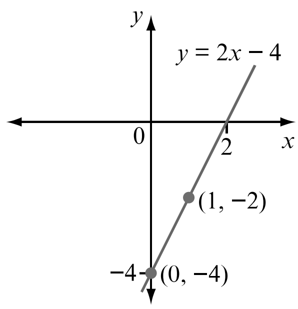
18 a $1 \le x \le 6$ b $-4 > x > -7$

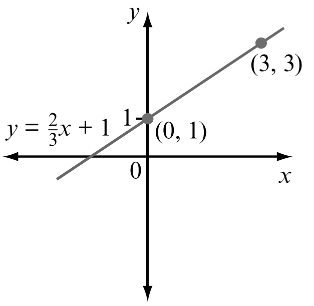
4C Sketching linear graphs

4C Start thinking!

1 The graph of a linear relationship is a straight line.
2 A straight line connects two points, and this will define the linear relationship.
3 a i $m = 4$, $c = 1$ ii $m = -4$, $c = 1$
iii $m = \frac{1}{4}$, $c = 1$
b i (0, 1) ii (0, 1) iii (0, 1)
c i 4 ii −4 iii $\frac{1}{4}$
d i rise = 4, run = 1 ii rise = −4, run = 1
iii rise = 1, run = 4
e i $y = 4x + 1$ matches graph C
ii $y = -4x + 1$ matches graph A
iii $y = \frac{1}{4}x + 1$ matches graph B
4 Starting at y-intercept, use rise and run to locate a second point. Ruling a straight line passing through these two points will represent the graph of the linear relationship.

Exercise 4C Sketching linear graphs

1 a


b


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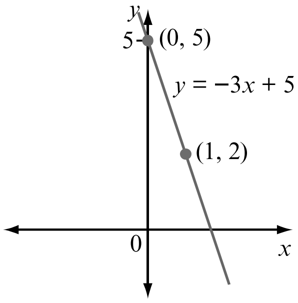

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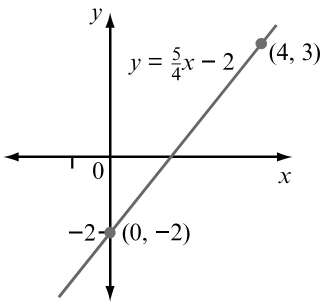

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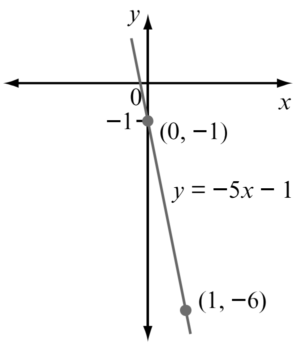

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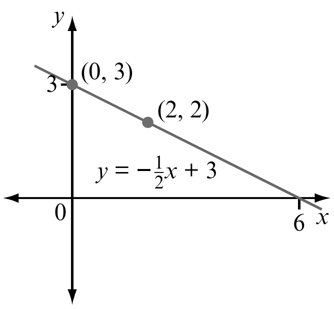

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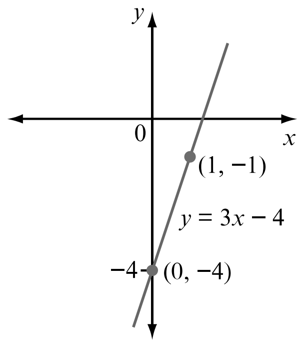

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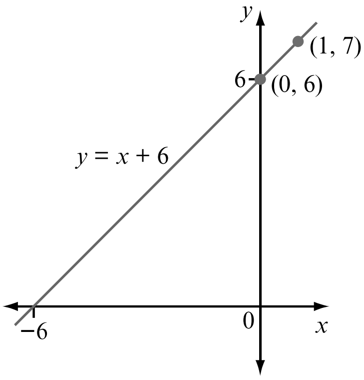
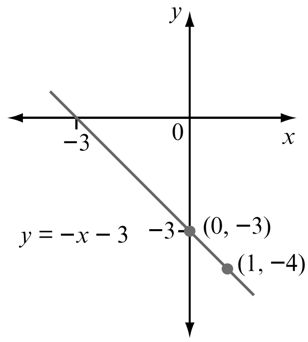
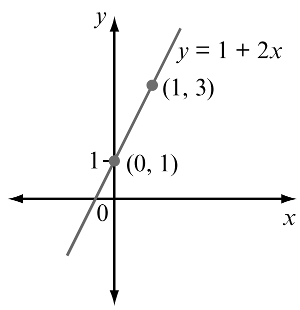


i


2 a $y = 2x + 1$, $m = 2$, $c = 1$
b $y = -3x + 4$, $m = -3$, $c = 4$
c $y = -\frac{3}{4}x - 3$, $m = -\frac{3}{4}$, $c = -3$
d $y = 3x - 6$, $m = 3$, $c = -6$
e $y = -2x + 6$, $m = -2$, $c = 6$
f $y = -5x - 2$, $m = -5$, $c = -2$
g $y = -3x + 2$, $m = -3$, $c = 2$
h $y = 4x - 5$, $m = 4$, $c = -5$
i $y = -2x + 3$, $m = -2$, $c = 3$
j $y = \frac{4}{5}x + 3$, $m = \frac{4}{5}$, $c = 3$
k $y = \frac{3}{2}x + 2$, $m = \frac{3}{2}$, $c = 2$
l $y = -\frac{1}{3}x - 2$, $m = -\frac{1}{3}$, $c = -2$

3 a


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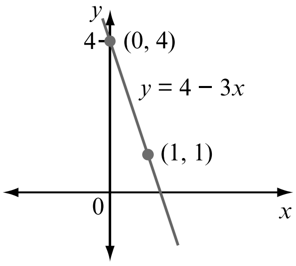

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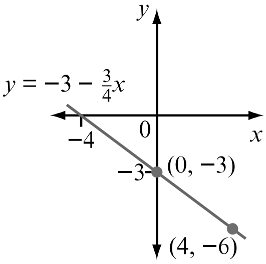

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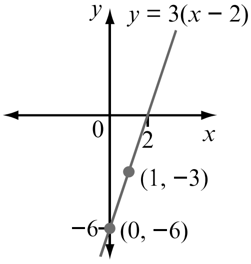

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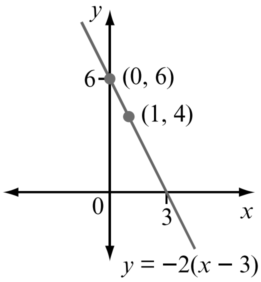

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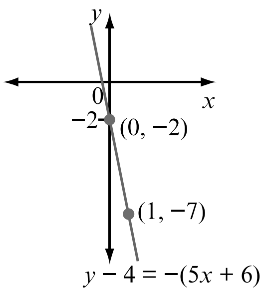

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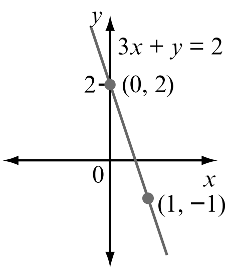

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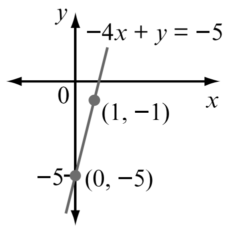

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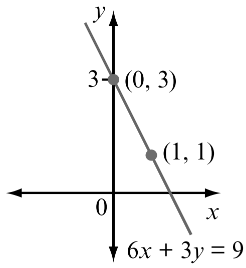

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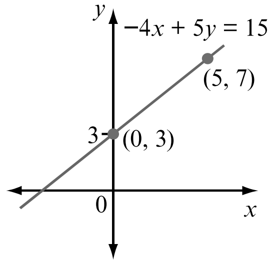

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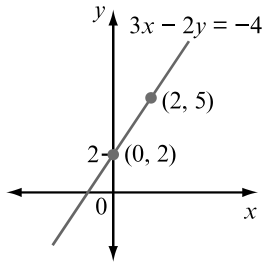

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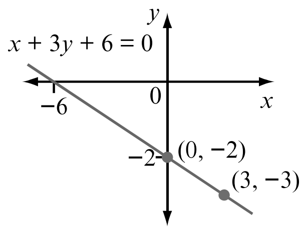
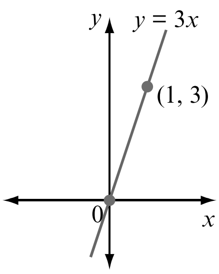


4 a $y = 3x + 0$

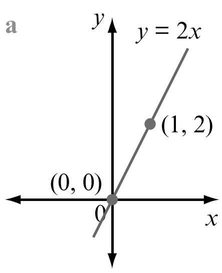
b **i** $m = 3$

ii $c = 0$

c



5 a



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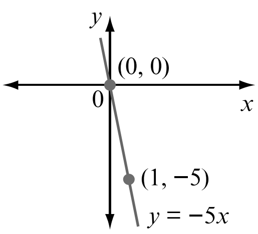

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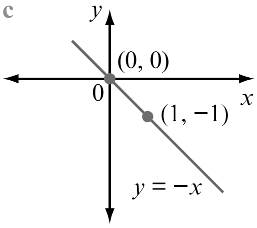

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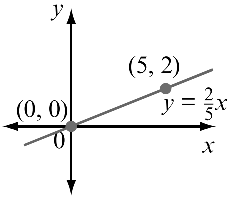
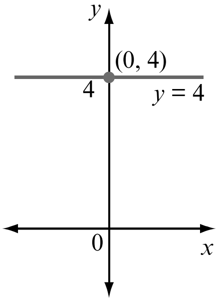


6 a $y = 0x + 4$

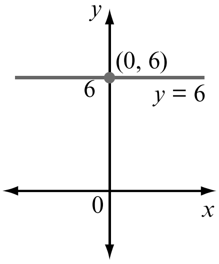
b **i** $m = 0$

ii $c = 4$

c



7 a



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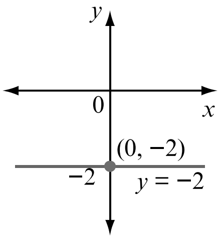

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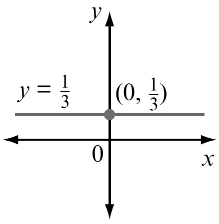

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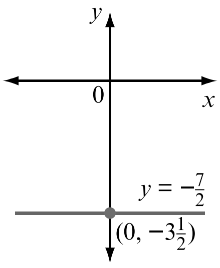
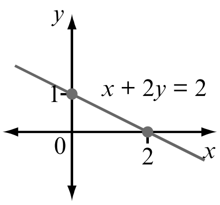


8 The rule $x = 4$ cannot be written in the form $y = mx + c$ since there is no y value specified.

9 a



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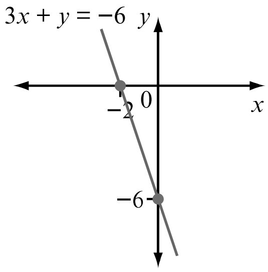

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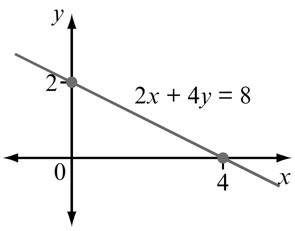

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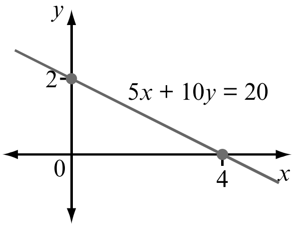

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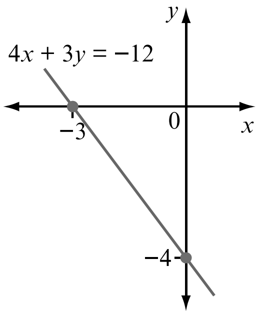

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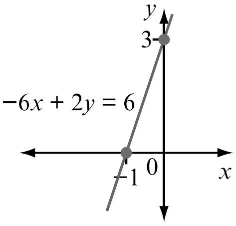

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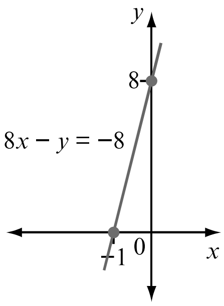

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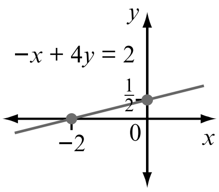

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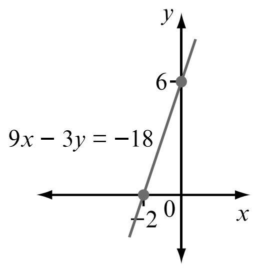

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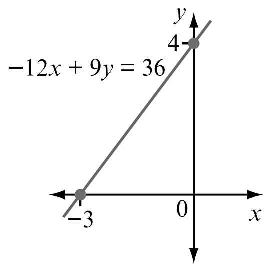

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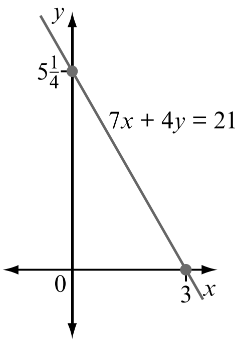

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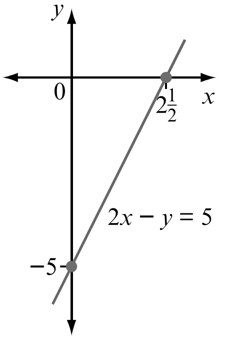
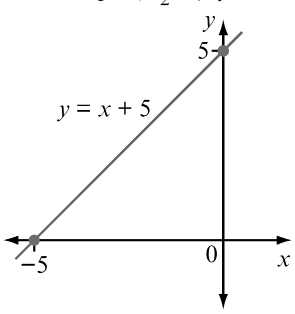


10 **a** x-intercept: $(-5, 0)$, y-intercept: $(0, 5)$
b x-intercept: $(4, 0)$, y-intercept: $(0, -4)$
c x-intercept: $(-2, 0)$, y-intercept: $(0, 6)$
d x-intercept: $(4, 0)$, y-intercept: $(0, -8)$
e x-intercept: $(2, 0)$, y-intercept: $(0, 2)$
f x-intercept: $(-\frac{1}{2}, 0)$, y-intercept: $(0, -3)$

11 **a**



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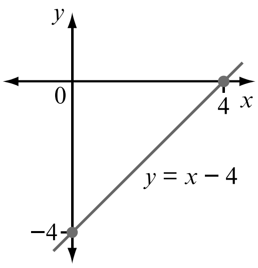
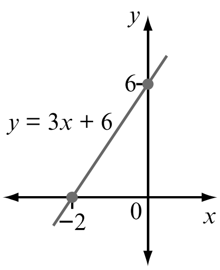


c



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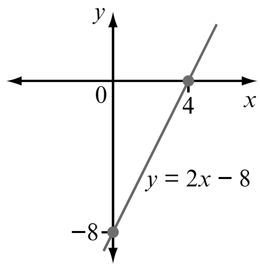

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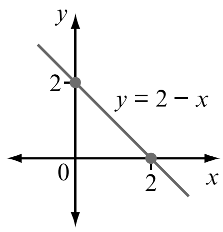

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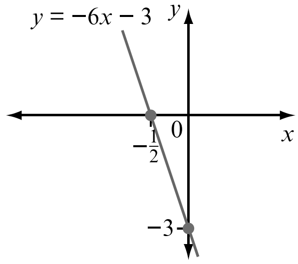
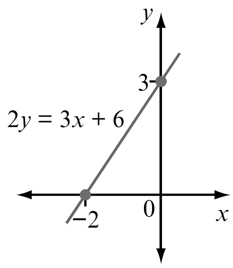
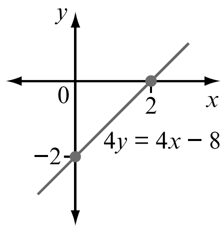


12 **a**



b



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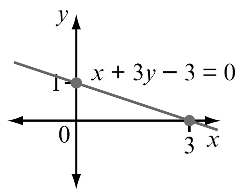
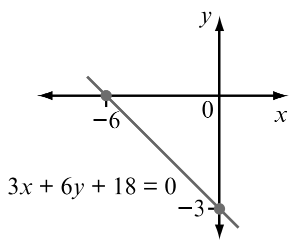
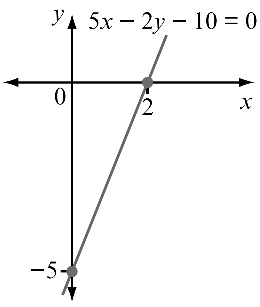
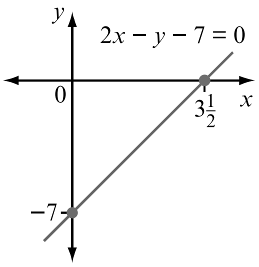
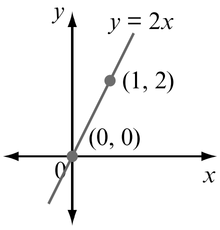


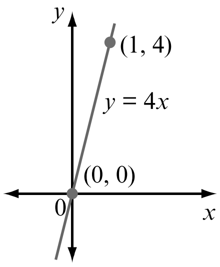
d

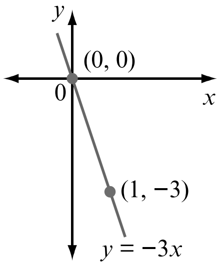


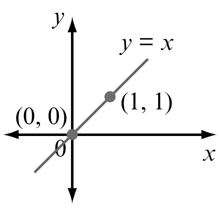
e 

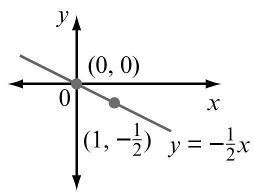
f 

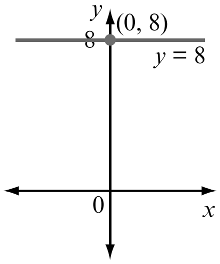
13 a i x-intercept: (0, 0) ii y-intercept: (0, 0)
b no; only have one point
c coordinates of one other point on the line
d $y = 2$ e (0, 0) and (1, 2)
f 

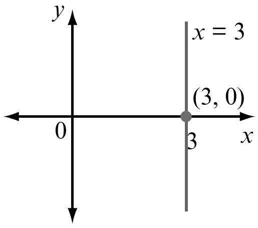
14 a 

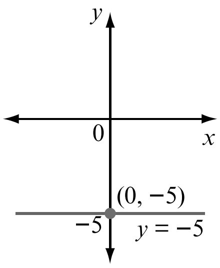
b 

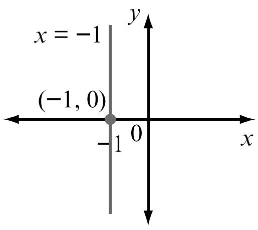
c 

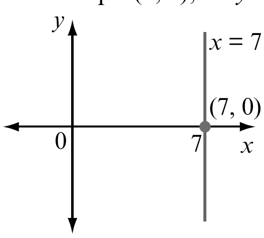
d 

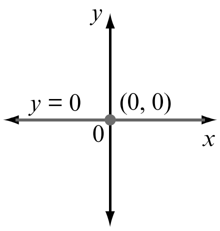
15 a i y-intercept: (0, 8), no x-intercept
ii 

b i x-intercept: (3, 0), no y-intercept
ii 

c i y-intercept: (0, −5), no x-intercept
ii 

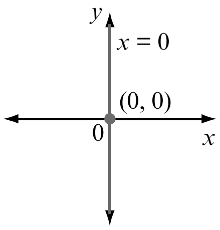
d i x-intercept: (−1, 0), no y-intercept
ii 

e i x-intercept: (7, 0), no y-intercept
ii 

f i y-intercept: (0, 0), no x-intercept
ii 

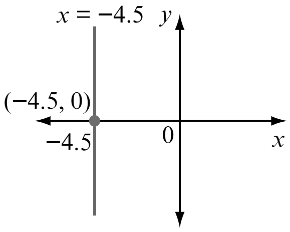
g **i** x-intercept: (0, 0), no y-intercept

ii



h **i** x-intercept: (−4.5, 0), no y-intercept

ii



16 A linear graph can have one or two axis intercepts (or perhaps lie along the x-axis or y-axis).

a x-intercept: (2, 0), y-intercept: (0, −4), two intercepts

b no x-intercept, y-intercept: (0, −7), one intercept

c x-intercept: (3, 0), y-intercept: (0, 2), two intercepts

d x-intercept: (0, 0), y-intercept: (0, 0), one intercept

e x-intercept: (10, 0), y-intercept: (0, −50), two intercepts

f x-intercept: (−5, 0), y-intercept: (0, 3), two intercepts

g x-intercept: (28, 0), no y-intercept, one intercept

h x-intercept: (6, 0), y-intercept: (0, −2), two intercepts

i no x-intercept, y-intercept: (0, 5.6), one intercept

17 **a** gradient–intercept method

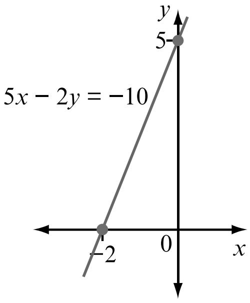
b x- and y-intercept method

c Find the coordinates of another point, and draw the straight line through that point and the origin.

d Locate the y-intercept, and draw a line through that point parallel to the x-axis.

e Locate the x-intercept, and draw a line through that point parallel to the y-axis.

18 **a**



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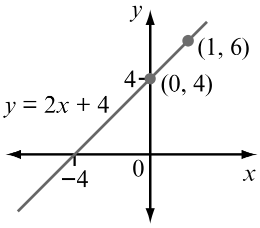

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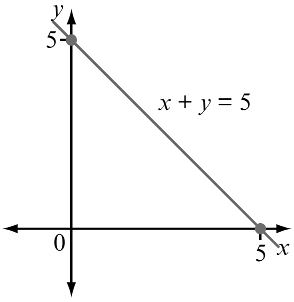

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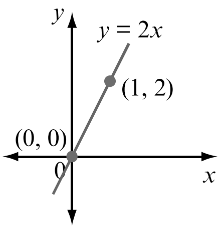

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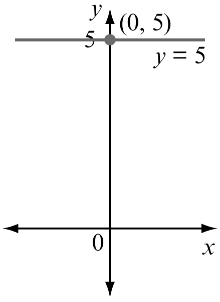

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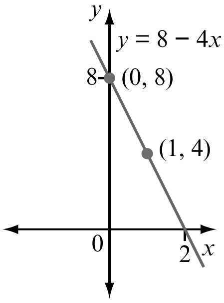

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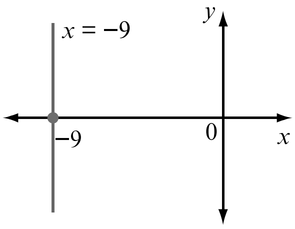

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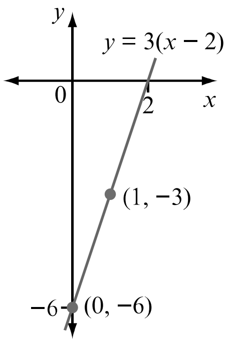

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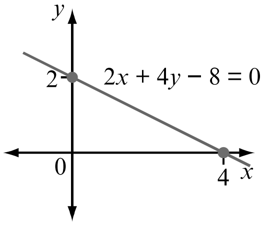

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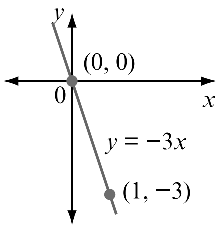

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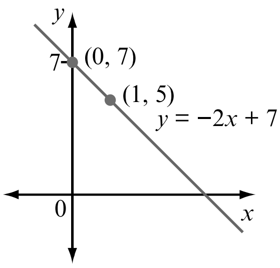

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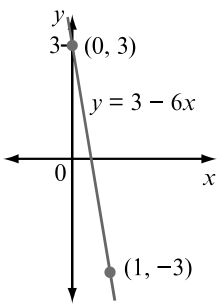
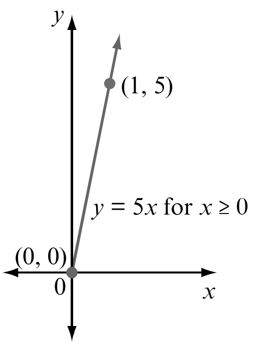


19 a


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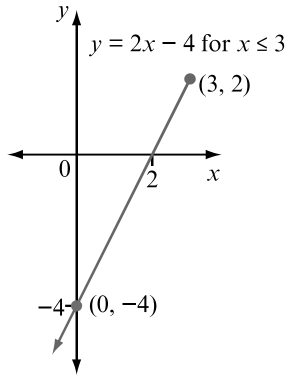

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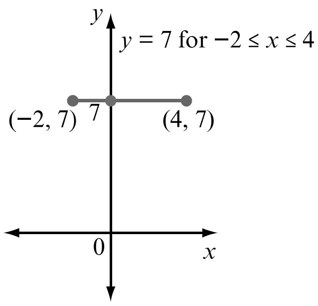

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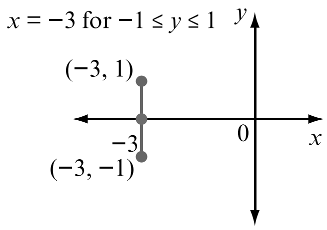

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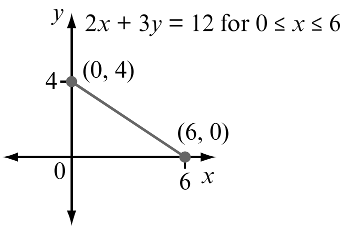

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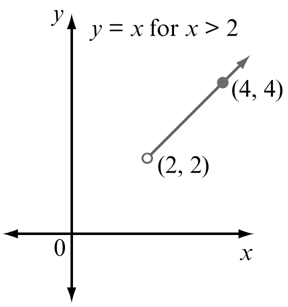

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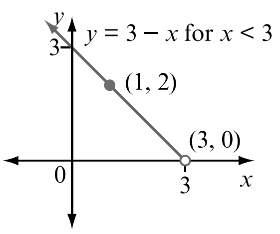

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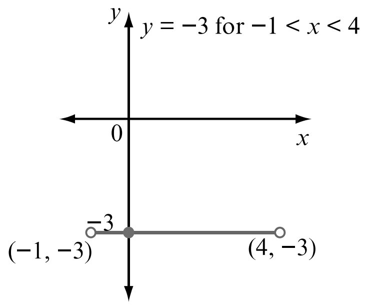

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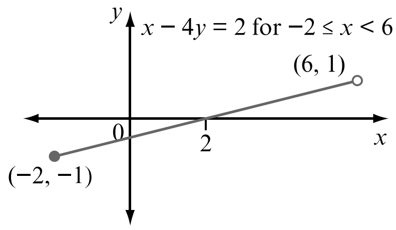

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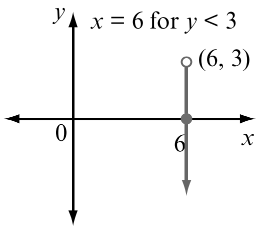
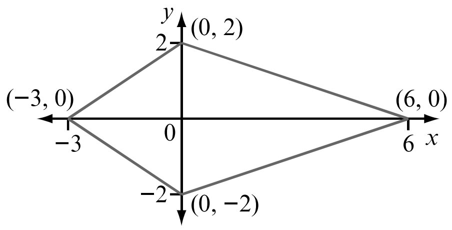
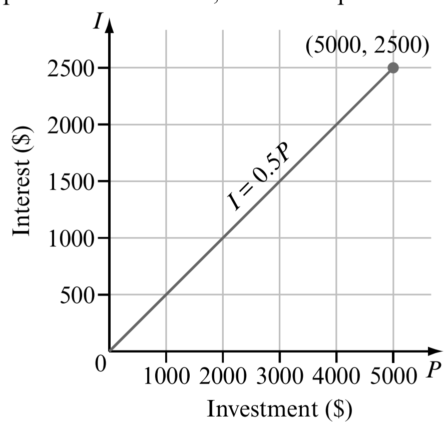
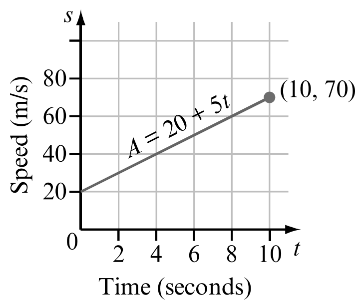


20 a


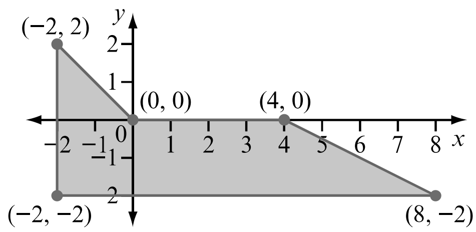
b kite
c i 19.9 units ii 18 square units
d coordinates of midpoints: (3, 1), (3, −1), (−1.5, −1) and (−1.5, 1)
e perimeter =13 units, area = 9 square units

21 a


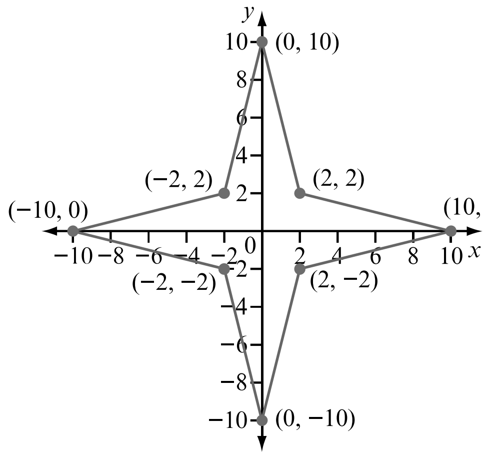
b i \$500 ii \$1250 iii \$2125
c i \$1500 ii \$2000 iii \$4500
d Gradient represents value of RT. This is the rate at which the investment grows over a 10-year period. $RT = 0.05 \times 10 = 0.5$.

22 a


b initial constant speed
c rate of increase in speed
d i 35 m/s ii 45 m/s iii 57.5 m/s
e i 126 km/h ii 162 km/h iii 207 km/h
f i 2 s ii 7 s iii 9 s
g 8 s h 252 km/h reached after 10 s

23 a


b 25.3 units c 18 square units

24 a


b 66 units c 80 square units
d $y = x$ e $y = -x$

4D Finding the rule for a linear relationship

4D Start thinking!

1 a Graph shows line drawn through two points on the Cartesian plane.
b Rule can be calculated from gradient and y-intercept.
c 3 d (0, 2) e $m = 3, c = 2$
f $y = 3x + 2$

2 c −2 d (0, 3) e $m = -2, c = 3$
f $y = -2x + 3$

3 General rule for straight line is $y = mx + c$. Gradient (m) is coefficient of x, and y-coordinate of y-intercept represents value of c.

4 a $y = 3x - 2$
b $y = -\frac{5}{3}x + 1$ or $5x + 3y = 3$